Test 5 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"Leila participated in a science exhibition at her school. She created a model demonstrating renewable energy sources. Her team explained solar panels, wind turbines, and hydroelectricity to visitors. Leila felt accomplished as many students showed interest and asked questions about sustainable energy."

a) What project did Leila present at the science exhibition?
b) Name three renewable energy sources mentioned in the passage.
c) How did Leila feel after presenting her project?
d) Rewrite the passage in your own words using four sentences.

2. Grammar & Vocabulary
a) Join the sentences with a suitable conjunction:
"The sun set quickly. The children continued playing in the park."

b) Identify all verbs in this sentence:
"The artist skillfully painted a mural on the school wall."

c) Correct the sentence:
"Neither the teacher nor the students was ready for the surprise test."

d) Convert to indirect speech:
"Ali said, 'I will participate in the debate competition tomorrow.'"

3. Writing
Write a short essay (120–150 words) on the topic: "Benefits and Drawbacks of Online Learning". Include examples and your personal viewpoint.

Section B: Mathematics

1. Algebra
a) Solve for y:
$3y + 7 = 2y + 15$

b) Factorize:
$x^2 – 14x + 45$

c) Simplify:
$6a^2b – 4ab^2 + 2a^2b$

2. Geometry & Mensuration
a) A triangle has a base of 20 cm and height of 8 cm. Calculate the area.
b) A cone has a radius of 4 cm and height of 9 cm. Find the volume.

3. Ratio, Proportion & Percentage
a) A jacket costs $120. It is sold at a 25% discount. Find the final price.
b) If 5 pens cost $15, find the cost of 18 pens.

4. Data Handling
Marks of 12 students in a social studies test: 14, 17, 16, 12, 15, 18, 17, 16, 13, 15, 14, 16.
a) Find the mean, median, and mode.
b) Represent the data using a frequency table.

Section C: Science

1. Biology
a) Explain the role of the human heart.
b) What is the difference between aerobic and anaerobic respiration?
c) List two differences between insects and mammals.

2. Physics
a) A bike increases its speed from 5 m/s to 25 m/s in 4 seconds. Calculate acceleration.
b) Define work and give an example.
c) Distinguish between reflection and refraction of light.

3. Chemistry
a) Identify elements, compounds, or mixtures:

- **Magnesium oxide**
- **Nitrogen**
- **Seawater**

b) Write a balanced chemical equation for the reaction of iron with hydrochloric acid.

4. Environmental Science
a) List two soil pollutants and their impact.
b) Suggest three practical ways to reduce plastic use at home.

Section D: Social Studies

1. History
a) Name two significant revolutions of the 20th century and their outcomes.
b) Explain the importance of the American Civil Rights Movement.
c) Identify the main countries in the Axis alliance during World War II.

2. Geography
a) Define equator and explain its impact on climate.
b) Name the largest desert in Africa and its location.
c) Explain two consequences of deforestation on wildlife.

3. Civics / Political Science
a) Differentiate between parliamentary and presidential systems of government.
b) State two duties of citizens in a democratic society.

4. Economics
a) Explain opportunity cost with an example.
b) Differentiate between direct and indirect taxes.

Section E: ICT / Computer Studies

1. Computer Basics
a) What is cloud computing? Give one example.
b) Name one input device and one output device.

2. Applications
a) Which spreadsheet function returns the minimum value from a dataset?
b) List two uses of word processing software in school.

3. Programming Basics
a) Define an array with an example in Python.
b) Write Python code to calculate the factorial of 5.

4. Cyber Awareness
a) Explain ransomware and how it can be prevented.
b) Mention two practices to protect personal data online.

Section F: Islamiyat

1. Quran & Hadith
a) Name two values emphasized in the Quran related to social behavior.
b) Narrate a Hadith that teaches patience.

2. Aqidah / Belief
a) List the five pillars of Islam.
b) Explain the importance of Tawheed in a Muslim's faith.

3. Fiqh / Worship
a) What is the significance of performing Hajj?
b) Describe the benefits of fasting in Ramadan for spiritual and physical health.

4. Islamic Ethics & Morality
a) How does Islam encourage justice in business?
b) Explain the importance of caring for animals and nature according to Islam.

Section G: Arts & Music

1. Visual Arts
a) Explain the use of shading in drawing.
b) Name two tertiary colors.

2. Music
a) Define tempo and dynamics.
b) Give an example of a wind instrument.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises to improve cardiovascular health.
b) List one indoor sport and one outdoor recreational activity.

2. Health & Hygiene
a) Why is mental health important for teenagers?
b) Mention two methods to prevent waterborne diseases.

3. Safety & Environment
a) State one rule for safe bicycle riding.